

# ReLU#

<https://pytorch.org/docs/stable/generated/torch.nn.ReLU.html>

## Description:

Applies the rectified linear unit function element-wise.  $\text{ReLU}(x) = \max(0, x)$

$\text{ReLU}(x) = (x)^+ = \max(0, x)$

inplace (bool) – can optionally do the operation in-place. Default: False

Input:  $(*)^{(*)}^{(*)}$ , where  $*^{**}$  means any number of dimensions. Output:  $(*)^{(*)}^{(*)}$ , same shape as the input.

## Function Signature

```
class torch.nn.ReLU(inplace=False)[source]#
```

Parameters

Shape:

```
extra_repr()[source]#
```

Return type

### Returns:

Tensor

## Code Examples

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```
>>> m = nn.ReLU()
>>> input = torch.randn(2)
>>> output = m(input)
```

An implementation of CReLU - <https://arxiv.org/abs/1603.05201>

```
>>> m = nn.ReLU()
>>> input = torch.randn(2).unsqueeze(0)
>>> output = torch.cat((m(input), m(-input)))
```

## Detailed Documentation

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### Documentation

Applies the rectified linear unit function element-wise.

$\text{ReLU}(x) = \max(0, x)$

`inplace (bool)` – can optionally do the operation in-place. Default: False

Input:  $(*)^{(*)} (*)$ , where  $***$  means any number of dimensions.

Output:  $(*)^{(*)}(*)$ , same shape as the input.

Return the extra representation of the module.

Runs the forward pass.

Applies the rectified linear unit function element-wise.

$\text{ReLU}(x) = (x) + \max(0, x)$   $\text{ReLU}(x) = (x) + \max(0, x)$

inplace (bool) – can optionally do the operation in-place. Default: False

Input:  $(*)^{(*)}(*)$ , where  $*^*$  means any number of dimensions.

Input:  $(*)^{(*)}(*)$ , where  $*^*$  means any number of dimensions.

Output:  $(*)^{(*)}(*)$ , same shape as the input.

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Return the extra representation of the module.

Runs the forward pass.